

(Please write your Enrollment Number)

Enrollment No. _____

REMID-TERM EXAMINATION
(Course Name: B.Tech) (Semester: I)
(November 2024) OFF LINE mode

Subject Code: BAI-101	Subject: PROGRAMMING WITH PYTHON
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
a)	Which of the following raises a TypeError in Python? If no TypeError occurs, write the output of each expression with an explanation. A. 5 * True B. 5 * "Spam" C. 5 * [1, 2] D. 5 * None E. C and D		CO1
b)	What is an infinite loop, and how can it occur in a while loop? Provide an example and explain how to avoid it.		CO1
c)	How do dictionaries handle duplicate keys? What happens if you assign a new value to an existing key?		CO2
d)	Write a program that creates a frozenset from a list and demonstrates that it is immutable by attempting to modify it. How can sets and frozensets be traversed? Provide examples.		CO2
UNIT I			
Q2	Attempt any two parts	(5*2=10)	CO Mapping
a)	Explain operator precedence and associativity in Python with examples. Evaluate the following expression. Explain step-by-step how operator precedence and associativity influence the result: result = 10 + 2 * 3 ** 2 // 4 - 5 Write a program to demonstrate the ternary conditional operator implemented in python.		CO1
b)	Write a Python program to read an odd-length string from the keyboard. Swap the parts before and after the middle three characters while keeping the middle three characters in place. Assume the string has at least 5 characters. Examples: "firstabclast1" → "last1abcfirsr" "abcdefghi" → "ghidefabcr"		CO1

c)	Given Python's ability to handle dynamic typing, explain the advantages and pitfalls of reusing identifiers for variables of different types within the same program. Provide a code snippet to demonstrate your explanation.		CO1
UNIT II			
Q3	Attempt any two parts	(5*2=10)	CO Mapping
a)	Twin primes are pairs of prime numbers that differ by 2. For example, (3, 5), (5, 7), and (11, 13) are twin primes. Write a Python program to read two positive integers, n and m, where $n < m$, from the keyboard. The program should print all unique twin primes between n and m (both inclusive). The output should be a list of tuples, where each tuple (a, b) represents a unique twin prime such that $n \leq a < b \leq m$. Input 1 15 Expected Output [(3, 5), (5, 7), (11, 13)]		CO2
b)	Read two integers from the keyboard, and write a Python program to print the number of unique digits that are common to both numbers. Example: For the input 287498 and 295424, the digits 2, 4, and 9 are common to both numbers. Therefore, the output should be 3.		CO2
c)	Write a Python script to process a list of customer orders, where each order is represented as a dictionary: <pre>orders = [{'id': 1, 'item': 'Laptop', 'quantity': 2, 'price': 1000}, {'id': 2, 'item': 'Phone', 'quantity': 5, 'price': 500}, {'id': 3, 'item': 'Tablet', 'quantity': 3, 'price': 300}]</pre> The program should: • Calculate the total value of each order. • Print a summary of all orders. • Identify the order with the highest total value.		CO2